Problem set #4

Due on December 7th by 12.30pm. Send solution to econ628vse@gmail.com
Answers must be typed. You can work in teams under the constraint: Max(Team Size)=2

1. (**Event Study**) Download the file "curfew.csv" from the `data` folder on the course website. This file contains data on the number of arrests of youth in a panel of cities enacting juvenile curfew laws in different years.

   a) Construct a variable D_{it} that equals one in the year that a city enacts a curfew law (e.g. when year==enacted).

   b) Run an event study of log juvenile arrests on curfew enactment with city and year effects. Be sure to "bin up" the endpoints and to cluster the standard errors by city. Impose a normalization in your event study coefficients so that all effects can be measured relative to the period before enactment.

   c) Make a plot of the event study coefficients with confidence intervals. What do you conclude about the impact of the program on arrests?

   d) Try changing the number of leads and lags in your event study. How do the coefficients change?

   e) Try adding city specific trends to the event study as controls. How do the coefficients change?

   f) Consider a region in the US as defined here `https://www.worldatlas.com/articles/the-regions-of-the-united-` `html`. Find the region of the 53 cities present in the dataset. Then, fit an event study regression model where you control for region by year fixed effects. How do the coefficients change?

   g) Now estimate a difference in differences model of the form (aka static DiD model)

$$y_{it} = \alpha_i + \lambda_t + \beta \tilde{D}_{it} + r_{it} \tag{1}$$

   where $\tilde{D}_{it} = \mathbf{1}\{t \geq t_i^*\}$ where t_i^* is the year of enactment of the curfew law for city i. How does β compare to your event study coefficients? In particular, do you find β to roughly correspond to an average of the event study coefficients post intervention that you estimated in part (b)? Give some intuition behind the patterns that you estimated – Note: it might be useful to answer this question to review the slides that discussed potential pitfalls of the static DiD model

2. (**CRE Event Study**) The event study design relies on potentially strong parametric assumptions. One approach to testing those assumptions is to use a Chamberlain (1982) style CRE approach. Consider the projection of the time invariant city effects α_i on the city curfew enactment dates e_i:

$$\alpha_i = \sum_{t=t_0}^{T} \eta_t 1\left[e_i = t\right] + v_i$$

   where v_i is an error term, t_0 is the first year of the panel, and T is the last year. We therefore have the following restricted reduced form for log arrests:

$$Y_{it} = \gamma_t + \sum_{k=0}^{T-t_0} \delta_k 1\left[e_i = t-k\right] + \sum_{l=t_0}^{T} \eta_l 1\left[e_i = l\right] + v_i + \varepsilon_{it} \tag{2}$$

where the $\{\delta_k\}$ give the dynamic causal effects of curfew laws and the $\{\eta_l\}$ represent endogeneity due to the omitted person effects.

a) Reshape the data into wide format so that there is a separate lnarrests variable for each year. To keep the exercise manageable, restrict the sample to years 1984-1990.

b) Over the sample period in question, there are only 5 event dates: $\{85, 87, 88, 89, 90\}$. Use Stata's "mvreg" command to estimate seven unrestricted cross sectional regressions of the form:

$$Y_{it} = \tilde{\pi}_t + \sum_{k \in \{85,87,88,89,90\}} \pi_{k,t} 1[e_i = k] + r_{it} \text{ for } t \in \{84, 85, 86, 87, 88, 89, 90\} \quad (3)$$

where the $\{\pi_{k,t}\}$ are the unrestricted reduced form regression coefficients and the $\{\tilde{\pi}_t\}$ are reduced form year specific intercepts.

c) Now test the linear restrictions on the $\{\pi_{k,t}\}$ implied by the model in (2). For example, one restriction is that $\pi_{k,t} = \pi_{k,t'} \; \forall \, (t, t') < k$. Enumerate all of the restrictions and test them jointly. Can you reject the model? What important inferential assumptions are you using in constructing your p-values?

3. (**Mass-Layoffs**) At this link I created for you a a computer generated pseudo matched employer-employee dataset `https://www.dropbox.com/s/0xppet65s9aaahm/fake_matched_employer_employee.csv?dl=1`. These data contain 15 million person-year observations so make sure to use a computer with sufficient memory. In these data, each person-year observation provides information on log daily wages and log earnings of an individual i in year t, characteristics of the individual (e.g. gender) and the identity of the employer ("firmid") of individual i in period t which we will denote via the function $J(i,t)$ discussed in class. Notice that this panel is unbalanced. Some workers might be observed in period t and period $t+2$ but not in period $t+1$. You can assume that such individual was unemployed in period $t+1$ and so did not receive any labor market earnings (i.e. you can set her earnings to zero).

a) Here you will use these administrative records to identify mass layoffs. A mass-layoff corresponds to an event in which firm j experiences a reduction in size of more than 30% from year t to year $t+1$. Create a variable $e^*_{J(i,t),t} = 1$ if firm $J(i,t)$ experienced a mass layoff event in year t. Focus just on firms that in year t had 50 or more workers. How many mass-layoff events did you find? (Note: make sure to count also cases where a firm closes from year t to year $t+1$, i.e. a firm identifier j is present in year t but disappears from the data in $t+1$, $t+2$, etc.

b) Now focus on set of workers that suffered the mass-layoff event. Create a variable, layoff_{it} equal to 1 if worker i suffered for the first time in her career a mass-layoff in year t, the employment history of the worker is such that $J(i,t+1) \neq J(i,t) = J(i,t-1) = J(i,t-2)$ and the age of the worker in year t is less than 50. Call the group of workers for which at some point in their observed work history $\text{layoff}_{it} = 1$ the "treated workers". Call the group of workers for which $\text{layoff}_{it} = 0$, $\forall t$ the group of "potential controls". If a worker experience more than one mass-layoff event, just consider her earliest mass-layoff event. Are the observable characteristics of these treated workers different from the remaining pool of workers?

(c Now perform an "hybrid" exact matching. Specifically, find for each treated worker who experienced a mass-layoff in year t whether there exists a matched control worker that in year t has the same

province, 2-digit sector (variable csc), age, tenure[1] gender and the log wage at $t-2$ is within a 0.05 band from the log wage of the treated worker (i.e. if treated worker has $LogWages_{it-2} = 4$ the matched control worker must have $LogWages_{it-2} \in (3.95, 4.05)$. In case where a control worker is matched with a treated worker that experienced a mass layoff in year t but also with a worker who experienced a mass layoff in year $t+s$, always keep the pair with the earliest mass-layoff year. Hint: take a look at the command calipmatch in Stata.

(d Save the sample of matched pairs. How many pairs did you successfully match? How many treated workers do you ended up dropping? Is the sample of matched treated workers different from the sample of all potential treated workers? How would you improve the matching?

(e Now "balanced" the matched sampled created above. For example, if worker i has entered the labor market in 1990 but has missing records in any $t > 1990$, fill the missing observations by setting her earnings equal to zero. Hint: take a look at the command "tsfill" in Stata.

(f Construct the event study dummies:

$$D_{it}^{k} = \mathbf{1}\{t = t_i^* + k\} \tag{4}$$

where t_i^* is the year in which worker i suffers the mass-layoff event (or in the case of a control worker, the year in which the matched treated worker suffers the mass-layoff event). Plot the distribution of t_i^*. Based on that plot, binned your event studies accordingly and justify your reasoning. Plot now the average earnings in event time the mass-layoff both for treated and control (or placebo) workers. Comment on the figure.

(g Estimate the following event study specification

$$\text{Earnings}_{it} = \alpha_i + \lambda_t + \sum_{k=a}^{b} D_{it}^{k}\beta_k + \sum_{k=a}^{b}(D_{it}^{k} \times T_i)\theta_k + u_{it} \tag{5}$$

where T_i is an indicator equal to 1 if worker i experienced in her career a mass layoff (a treated worker) and a and b were chosen by you in the previous part. How do you interpret the coefficients β_k? What about θ_k? Plot the estimated $\{\theta_k\}$ and discuss the normalization that you imposed to estimate this model. How did you cluster the standard errors? Why?

(h Restimate the equation above but this time dropping the control workers. Do the coefficients change? Interpret the differences between the two models.

(l Having learned of the potential pitfalls of two-way fixed effects model in the presence of treatment effects heterogeneity, you decided to run a more sophisticated model where you estimate equation (5) separately by t_i^* (adding back the matched control workers in the estimation sample). Compute a weighted average of the event-study coefficients that look as follows

$$\dot{\theta}_k = \sum_{t^*} w_{t^*}\theta_k^{t^*} \tag{6}$$

where w_t^* is the fraction of treated workers who experienced a mass-layoff in year t^* and $\theta_k^{t^*}$ is the event-study coefficient obtained by running (5) just for cohort t_i^*. How does a plot of the event study

[1] $Tenure_{it}$ measures the amount of years that individual i has spent with her current employer up to year t (e.g. if Raffa started working for UBC in 2018 and he is still employed by UBC in t=2021, then $Tenure = 3$.

coefficients $\dot{\theta}_k$ compare to the original θ_k that you got from the (not saturated) event-study model from equation (5)? Comment on the potential pitfalls of two-way fixed effects model in the presence of (cohort-based) treatment effects heterogeneity.

(ll) Now re-estimate the event-study specification model in equation (2) using the methodology proposed by Borusyak, Jaravel, Spiess (2022-BJS henceforth). Create a plot where you overlay the effects on earnings obtained from the two-way fixed effects OLS specification described in part (g), part (h), the ones obtained in part (l) and the estimates from BJS. What do you conclude about the biases due to treatment effects heterogeneity in this application?